# Assignment 4

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**Aim**

To design and build an interactive weather dashboard using HTML, CSS, JavaScript, and Chart.js for visualizing weather data of different cities.

**Objective**

1. To develop a responsive layout using HTML and CSS.
2. To use JavaScript for updating content dynamically.
3. To visualize weather data using the Chart.js library.
4. To present temperature, humidity, and rainfall data in graphical form.
5. To update charts dynamically based on the city selected by the user.

**Theory**

A dashboard is a visual interface that presents important information in an easy-to-understand format. In this project, a weather dashboard is created using front-end technologies to display weather data interactively.

**1. HTML**

HTML is used to build the structure of the webpage.

- Elements like <select> and <button> are used for user input.
- <canvas> elements are used to display charts.
-  elements help organize the layout into sections or cards.

**2. CSS**

CSS is used to style and design the dashboard.

- Grid layout (display: grid) helps create a responsive structure.
- Cards are designed using properties like padding, border-radius, and box-shadow.
- Background colors and font styles improve the overall appearance.

**3. JavaScript**

JavaScript makes the dashboard interactive.

- Weather data is stored in an object called weatherData.
- The createCharts() function generates charts based on the selected city.
- The updateDashboard() function updates the data when the user selects a different city.

- Old charts are removed before creating new ones to prevent duplication.

**4. Chart.js**

Chart.js is a JavaScript library used for creating charts and graphs.

- A line chart is used to show temperature trends.
- Bar charts are used to display humidity and rainfall data.
- Data is plotted using labels (such as days of the week) and datasets.

**Output**

**Conclusion**

The interactive weather dashboard was successfully developed using HTML, CSS, JavaScript, and Chart.js. The experiment demonstrated how to create responsive layouts, implement dynamic data updates, and visualize information using charts. This practical helped in understanding front-end interactivity and data visualization techniques used in modern web applications.